

OpenAI Math Digest

CATALOGUE 038

Fujita's freeness conjecture

Catalogue guide & one proof overview

Independent reading guide · English edition

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Contents

The catalogue guide is followed by the article. The contents and citations are clickable. Wide tables are arranged as entries for print; the claims and source references are retained.

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Original-manuscript snapshot: `fd4aeeb2ee4fc729c18d98444fed42fd0529eeeb`

Catalogue 038 · Guide

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CATALOGUE 038

Fujita's freeness conjecture

This catalogue concerns global generation of adjoint bundles on smooth projective complex varieties at the sharp dimension-dependent exponent. The official catalogue contains one manuscript.

The paper in official catalogue order

These summaries give the principal scope; consult the cited original results for the complete hypotheses and statements.

Catalogue 038 · 2026-10-10 checked source snapshot

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01

Manuscript and overview

Fujita's freeness conjecture →

Short name: Fujita freeness

2026-09-23 · 25 pages

Original paper (GitHub) ↗

Principal claim

For a smooth connected projective complex n -fold and ample L , $K_X + mL$ is globally generated for every integer $m \geq n+1$.

Theorem 1.1 · p. 1; Corollary 6.3 · p. 23 ↗

Role and available overview

Exponential minimization, uniform restriction-image estimates, and lifting from local discrepancies. The overview covers the full proof, point centers, the equality divisor, and every higher exponent.

Read the proof overview →

Connections to external results

Start with the [complete proof route](#), then follow [exclusion of positive-dimensional centers](#), [isolation of the equality divisor](#), and [extension to every exponent](#). The last step uses a product with projective space and does not assume global generation of L itself.

The [input list](#) distinguishes direct inputs from semicontinuity, vanishing, Brunn–Minkowski and SNC resolution from the methodological reference to Fujita's local lifting argument. All cross-catalogue dependencies remain uninvestigated.

Sources and checking scope

The official subject is **Algebraic and complex geometry**. The source is the research catalogue dated October 6, 2026, at fixed commit `fd4aeeb2ee4fc729c18d98444fed42fd0529eeeb`.

OpenAI Research Catalog · Catalogue 038 · p. 5

The manuscript is by OpenAI, dated September 23, 2026, and has 25 pages including references. The article and original state the hypotheses and quantifiers explicitly.

Theorem 1.1 · p. 1; Corollary 6.3 · p. 23

The proof text and the statements and application sites of direct external inputs were compared. This is not independent verification of the entire proof. The [article's closing scope statement](#) identifies the external proofs and technical estimates not subjected to complete independent reconstruction or formal verification.

038 / 01

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CATALOGUE 038 · PAPER 01 / 1 Overview

Fujita's freeness conjecture

Exponential minimization isolates a point center; local discrepancies lift an adjoint section

The manuscript claims Fujita's freeness conjecture in every dimension for an arbitrary ample line bundle. The core argument combines exponential minimization arising from a hypothetical base point with a tangential first variation at a positive-dimensional center. A restriction-space estimate independent of the degree and singularities of the center isolates a point without losing the sharp exponent.

Manuscript author: OpenAI · 2026-09-23 · 25 pages, including references

[Read the manuscript on GitHub ↗](#)

1. Main results

Theorem 1.1 — Global generation at the sharp exponent

Let X be a smooth connected projective complex variety of dimension $n \geq 1$, and let L be an ample line bundle on X . Then

$$P := K_X + (n + 1)L$$

is globally generated. Global generation of L itself is not assumed. For $X = \mathbf{P}^n$, $L = \mathcal{O}_{\mathbf{P}^n}(1)$, the bundle P is trivial whereas $K_X + nL$ has no nonzero section, so the exponent is sharp. The statement does not include separation of points or tangent directions, or very ampleness. For disconnected varieties it applies componentwise; dimension zero is immediate.

Theorem 1.1 · p. 1

Corollary 6.3 — Every higher exponent

Under the hypotheses of Theorem 1.1, $K_X + mL$ is globally generated for **every integer** $m \geq n + 1$.

Corollary 6.3 · p. 23

References on the arrows

Unprefixed theorem, lemma, section and equation numbers refer to [Fujita freeness](#). External inputs use the following keys.

- **DK** Jean-Pierre Demailly; János Kollár — Singularity exponents: Semicontinuity of singularity exponents, used in threshold and flag-incidence loci. [arXiv:math/9910118v2](#), 2000-05-01.
- **KV** Osamu Fujino — Kawamata–Viehweg vanishing: Vanishing for a nef and big remainder and lifting of sections. Lecture note v1.04, 2009-07-22.
- **BM** Richard J. Gardner — Brunn–Minkowski: Growth of sums of initial-exponent sets forced by restricted sections. *Bulletin AMS* 39 (2002), 355–405; published version.
- **FL** Takao Fujita — Adjoint lifting: Methodological source for local discrepancy and lifting; its threefold main theorem is not applied. [arXiv:alg-geom/9311013v1](#), 1993-11-30.
- **RES** Jarosław Włodarczyk — SNC resolution: Resolution and principalization preserving an existing SNC boundary. *JAMS* 18 (2005), 779–822; published version.

Each citation gives the result and pages. When printed pagination differs from the PDF position, both are shown. GitHub previews do not automatically jump to the cited page.

2. Theorem 1.1 — Proof overview of the main theorem

Proof target : [Theorem 1.1](#) · p. 1

Proof route. Suppose that P is not generated at a closed point x . A strict ordinary-order estimate produces a minimization problem in which both the basis and the monomial order vary. After [excluding positive-dimensional centers](#) and [isolating the discrepancy-equality divisor](#), [vanishing](#) supplies a section nonzero at x . Finally, [application to a product](#) gives Corollary 6.3.

Put $R_j = H^0(X, jL)$ and $N_j = \dim R_j$. The test v is a nonzero monomial order with nonnegative real weights on a smooth birational model, in SNC coordinates adapted to the relative canonical divisor. If E_i are its positive-weight components with weights u_i , then $A(v) = \sum_i u_i A(E_i)$, where $A(E_i) = 1 + \operatorname{ord}_{E_i} K_{Y/X}$. The center $c_X(v)$ is the closed center, and the constraint is $x \in c_X(v)$. Positive rescaling changes orders and discrepancy by the same factor.

§2.1; Lemmas 2.1–2.2 · pp. 3–4

1 Assume a base point; obtain strict excess

$$P = K_X + (n+1)L, \quad x \in \text{Bs } |P|$$

$$\liminf_j \frac{1}{jN_j} \sum_{\gamma \in H_j} |\gamma|_1 > \frac{n}{n+1}$$

Equality would permit lifting; closed flag loci and SNC retraction give attainment.

[Proposition 3.2 · p. 6](#)

[Lemmas 4.2–4.3; Proposition 4.4 · pp. 10–12](#)

2 Minimize over bases and weighted orders

$$F_{m,t} = \frac{1}{N_m} \sum_i e^{A(v)/t - v(s_i)/m}$$

$$0 < c \leq f_{m,t}(x) \leq 1 - \delta, \quad 0 < a_* \leq A(v) \leq a^*$$

Compare tangential first variation with uniform restriction-image growth.

[Proposition 5.1 · p. 13](#)

3 Force every minimizing center to be a point

$$t < n+1 \text{ fixed}, \quad c_X(v) = \{x\}$$

For every minimizing pair in arbitrarily large degrees

Rationalize support inequalities and the strict budget; mix in an ideal minimum.

[§6.1; Lemma 6.1; \(6.1\)–\(6.5\) · pp. 19–20](#)

[Lemma 6.2 · pp. 21–22](#)

4 Isolate the equality divisor using rational data

$$S \neq 0, \quad \rho(S) = \{x\}, \quad \text{coeff}_F B = A(F) \quad (F \subset S)$$

Strict inequality on other adjacent components; nef and big remainder.

Lift a nonzero fiber value on S ; descend the exceptional positive correction.

[Kawamata–Viehweg vanishing \[KV\] · Theorem 0.1 · p. 1](#)

[§6.3; \(6.13\)–\(6.15\) · pp. 22–23](#)

[Lemma 2.5 · p. 6](#)

5 Lift and descend to remove the base point

$$H^1(T, \mathcal{O}_T(D)) = 0 \implies s \in H^0(X, P), \quad s(x) \neq 0$$

Contradicts failure at any point; proves Theorem 1.1

From a hypothetical base point through minimization and isolation to a nonzero section

[Open figure ↗](#) [TeX source ↓](#)

1. Lifting in the equality case makes the mean order strictly larger

Let H_j be the initial-exponent set for the Taylor monomial order comparing ordinary degree first. From $\#H_j = N_j$ and the Hilbert asymptotic, filling lattice points in increasing total degree gives

$$\liminf_{j \rightarrow \infty} \frac{1}{jN_j} \sum_{\gamma \in H_j} |\gamma|_1 \geq (L^n)^{1/n} \frac{n}{n+1}.$$

If the bound with right-hand side $n/(n+1)$ were an equality under failure of generation, then $L^n = 1$, and the normalized initial exponents would eventually meet every fixed small ball in the interior of the simplex. Choosing $n < t' < n+1$, the manuscript forms products of sections whose leading forms define an ideal with threshold greater than $1/h$ on the exceptional divisor E of the point blowup. Degeneration to monomials and semicontinuity from Singularity exponents [DK] transfer this strict inequality back to the original leading forms.

After resolving the ideal, the remaining positivity is $(n+1-t')L$. Theorem 2.4 kills H^1 , a nonzero fiber value on E lifts, and Lemma 2.5 descends it to a section nonzero at x . This contradicts failure of generation. Thus Proposition 3.2 gives the **strict** inequality $\liminf > n/(n+1)$.

Proposition 3.2 · pp. 6–9

Singularity exponents [DK] · Theorem 3.1; Lemma 3.2 · pp. 14–15

2. Fix a basis through a flag before taking a minimum on an SNC model

For an ordered degree- m basis $\mathbf{s} = (s_i)$ and $t > 0$, set

$$F_{m,t}(v, \mathbf{s}) = \frac{1}{N_m} \sum_i \exp \left(\frac{A(v)}{t} - \frac{v(s_i)}{m} \right), \quad f_{m,t}(x) = \inf_{x \in c_X(v), \mathbf{s}} F_{m,t}(v, \mathbf{s}).$$

Proposition 3.2 makes the derivative for $v = \lambda \text{ord}_x$ negative at $\lambda = 0$. Lemma 4.2 gives $0 < c \leq f_{m,t}(x) \leq 1 - \delta$ for a common interval immediately below $n+1$ and all sufficiently large m . On the relevant sublevel set it also gives $0 < a_* \leq A(v) \leq a^* < \infty$. The upper bound uses a supply of low-order jets forcing a fixed proportion of every basis to have low order.

(4.1) · p. 9

The proof does not compactify all bases or all models at once. Rational lower bounds on normalized section orders become a threshold condition for a sum of powers of base ideals. Lemma 2.3 gives closed loci in the complete flag variety. Choose a flag in the nested intersection and a basis adapted to it; only then resolve this basis's divisors on one model. SNC retraction preserves section orders and does not increase discrepancy, while normalized weights lie in a compact simplex. Weights may vanish in the limit, but the center then enlarges and still contains x . This yields the minimizing pair of Proposition 4.4.

Lemmas 4.2–4.3; Proposition 4.4 · pp. 10–12

SNC resolution [RES] · Theorem 1.0.1 · print p. 781 / PDF p. 3; Definition 2.1.3 · print p. 783 / PDF p. 5; Theorem 2.4.1 · print p. 786 / PDF p. 8

3. Force every minimizing center to be a point and rationalize support data

Proposition 5.1 fixes one $t < n + 1$ for which, in arbitrarily large degrees, **every** minimizing pair has center $\{x\}$. This universal assertion is needed next: another stratum realizing the same normalized section-order profile and discrepancy would give the same minimum.

On an SNC model of $D_i = \text{div}(s_i)$, normalized profiles from the strata form finitely many rational polytopes. The gradient of the objective gives positive supporting coefficients b_i , and differentiation in the scaling direction gives $m \sum_i b_i = t < n + 1$. Lemma 6.1 rationalizes the profile and supporting coefficients without changing membership in the polytopes or losing this strict budget. The ideal perturbation of Lemma 6.2 then isolates a nonempty reduced discrepancy-equality divisor S over x .

Proposition 5.1 · p. 13

§6.1; Lemma 6.1; (6.1)–(6.5) · pp. 19–20

Lemma 6.2 · pp. 21–22

4. A local coefficient comparison gives a section nonzero at the original point

On a resolution $\rho : T \rightarrow X$, the effective rational SNC divisor B leaves $(n + 1)\rho^*L - B$ nef and big. Put $D = K_T + (n + 1)\rho^*L - \lfloor B \rfloor$; then $D - (K_T + \{B\})$ is this remainder. Kawamata–Viehweg vanishing [KV] gives $H^1(T, \mathcal{O}_T(D)) = 0$, hence a surjection

$$H^0(T, \mathcal{O}_T(D + S)) \longrightarrow H^0(S, \mathcal{O}_S(D + S)).$$

The comparison divisor is

$$C = D + S - \rho^*P = K_{T/X} - \lfloor B \rfloor + S.$$

By Lemma 6.2, its coefficients are zero on S and nonnegative on other components meeting S . Thus C is effective near S and has no component of S in its support. Since S is reduced and all its components map to x , one has $\rho^*P|_S = P|_x \otimes \mathcal{O}_S$. Multiplying a nonzero fiber value by the canonical rational section of C gives a section nonzero at the generic point of every component of S . It lifts by the displayed surjection.

On nonexceptional components the coefficients of C are nonpositive, so its positive part is exceptional. Lemma 2.5 descends the lift to X with $s(x) \neq 0$. For $n = 1$, a component of S may be the point x itself, and the same coefficient comparison applies. The assumed failure at any point has been contradicted, completing the global-generation conclusion of Theorem 1.1. Only the comparison near S is needed: neither global log canonicity of B nor global effectivity of C is required.

§6.3; (6.13)–(6.15) · pp. 22–23

Theorem 2.4 · p. 5 · Lemma 2.5 · p. 6

3. Proof of Proposition 5.1 — Excluding positive-dimensional centers

Proof target : Proposition 5.1 · p. 13

Proposition 5.1 — Point centers for all minimizing pairs

For X, L, n as in Theorem 1.1 and $P = K_X + (n + 1)L$, suppose that P is not generated at a closed point x . For the function $f_{m,t}(x)$ defined above, there is a real number $0 < t < n + 1$ such that, in arbitrarily large integer degrees m , the infimum is attained and every monomial test and basis attaining it has center exactly $\{x\}$.

Proposition 5.1 · p. 13

Proof route. If a positive-dimensional center W exists, minimality bounds a weighted tangential moment from above. Uniform growth of restricted sections forces a strictly larger lower bound for the same moment. Constants independent of centers and models, and the order of limits, connect the two estimates.

1 Assume a positive-dimensional center

$$\dim W = d > 0, \quad I_W(h) = \dim \operatorname{im}(R_h \rightarrow H^0(W, hL|_W))$$

$$(d!I_W(h))^{1/d} \geq h - C_d \quad (\text{uniform in } W)$$

Specialization at a very general point y makes the right derivative nonnegative.

[Lemma 4.5 · pp. 12–13](#)

[§5.2; \(5.4\)–\(5.10\) · pp. 14–16](#)

2 Bound the tangential moment from above

$$A(v_\epsilon) = A(v) + d\epsilon + O(\epsilon^2)$$

$$T_m = \frac{\sum |\beta|_1 e^{-\ell(p)/m}}{mQ_m} \leq \frac{d}{t}$$

Use uniform restriction images and sums of cubes; weight the slice growth.

[Lemma 5.2; \(5.1\)–\(5.3\) · pp. 13–14](#)

[Brunn–Minkowski \[BM\] · Theorem 4.1 · print p. 362 / PDF p. 8](#)

[§5.3; \(5.11\)–\(5.16\) · pp. 16–17](#)

3 Bound the same moment from below

$$b_p(j+h) \geq b_p(j) + h - C'_d$$

$$\liminf T_m \geq dI_*, \quad I_* = \int_0^1 s^n e^{(1-s)a_*/(n+1)} ds$$

Fix t with $I_* > 1/t$ first; send m to infinity, then the partition size.

[§5.4; \(5.17\)–\(5.21\) · pp. 17–19](#)

4 Contradict at fixed t ; all minimizing centers are points

$$\frac{d}{t} \geq \liminf T_m \geq dI_* > \frac{d}{t}$$

One t ; arbitrarily large m ; every minimizing pair

Uniform restriction-space growth and tangential first variation bound the same weighted moment

[Open figure ↗](#) [TeX source ↓](#)

1. Separate the restriction-image bound from the center's degree and singularities

Fix a very ample multiple kL . For any integral d -dimensional subvariety W , put

$I_W(h) = \dim \operatorname{im}(R_h \rightarrow H^0(W, hL|_W))$. Lemma 5.2 gives

$$(d!I_W(h))^{1/d} \geq h - C_d \quad (h \geq H_d)$$

with C_d, H_d independent of W . Surjectivity of restriction is not used. A general linear projection $W \rightarrow \mathbf{P}^d$ has degree $D = k^d(L^d \cdot W) \geq D_0 = k^d$. Bounded-degree ambient forms separate D_0 points of its geometric generic fiber and yield D_0 elements independent over $\mathbf{C}(\mathbf{P}^d)$.

Multiplication by monomials from the base gives $I_W(qk) \geq D_0 \binom{q-D_0+d}{d}$. Globally generated multiples handle the finitely many residue classes and cover all degrees. The argument uses the fixed number D_0 of points, not an upper bound on D .

Lemma 5.2; (5.1)–(5.3) · pp. 13–14

2. Use specialization to make the tangential derivative nonnegative

Suppose a minimizing pair has center W of dimension $d > 0$. Lemma 4.5 gives $f_{m,t}(y) \geq f_{m,t}(x)$ at a very general smooth point $y \in W$. On the stratum above it, split coordinates into the old positive-weight directions, fiber directions and tangential directions from W . Write initial exponents as (p, β) and the old weight as $\ell(p)$. Replacing the basis by one adapted to this initial order preserves minimality.

Give weight ϵ to the d tangential coordinates and ϵ^2 to the remaining fiber coordinates. For each fixed basis, $A(v_\epsilon) = A(v) + d\epsilon + O(\epsilon^2)$ and $v_\epsilon(s) = \ell(p) + \epsilon|\beta|_1 + O(\epsilon^2)$. The perturbed center is y , so specialization is what makes $F(v_\epsilon, \mathbf{s}) \geq f_{m,t}(y) \geq F(v, \mathbf{s})$ available. Consequently

$$T_m := \frac{\sum_{(p,\beta) \in \Gamma_m} |\beta|_1 e^{-\ell(p)/m}}{mQ_m} \leq \frac{d}{t}, \quad Q_j := \sum_{(p,\beta) \in \Gamma_j} e^{-\ell(p)/m}.$$

The denominator in the exponential defining Q_j is the reference degree m , not the auxiliary degree j . Perturbation and differentiation are performed at fixed m ; no choice of ϵ uniform in all degrees is required.

Lemma 4.5 · pp. 12–13 · §5.2; (5.4)–(5.10) · pp. 14–16

3. Weight the growth of sumsets to contradict the upper bound

For a fixed transverse exponent p , consider the slice $G_p(j)$. Restricted sections supply enough points in $G_0(h)$, and in the fixed degree $h_0 = kd$ one also has $\{0, 1\}^d \subset G_0(h_0)$. Thicken finite sets by unit cubes and apply Brunn–Minkowski [BM]. For nonempty slices, $b_p(j) = (d! \#G_p(j))^{1/d}$ satisfies

$$b_p(j+h) \geq b_p(j) + h - C'_d,$$

where C'_d is independent of the center, test and j, p . Weight this inequality by $e^{-\ell(p)/m}$. Convexity of the $(d+1)$ st power and a simplex first-moment estimate turn it into a lower bound for T_m .

§5.3; (5.11)–(5.16) · pp. 16–17 · Brunn–Minkowski [BM] · Theorem 4.1 · print p. 362 / PDF p. 8

The uniform $a_* > 0$ from Lemma 4.2 gives

$$I_* := \int_0^1 s^n \exp\left(\frac{(1-s)a_*}{n+1}\right) ds > \frac{1}{n+1}.$$

Fix $t < n + 1$ once so that $I_* > 1/t$. Choose a subsequence along which $f_{m,t}(x)$ approaches its positive liminf. If positive-dimensional centers occur infinitely often, pass further to a fixed dimension d . At auxiliary degrees $j_i = \lfloor mi/l \rfloor$, the rescaled test $(j_i/m)v$ is admissible at x , so specialization at y is not needed in every auxiliary degree. First let $m \rightarrow \infty$ for fixed l , and only then let $l \rightarrow \infty$. The result is

$$\liminf T_m \geq dI_* > \frac{d}{t},$$

contradicting the first-variation bound. Far enough along this subsequence, no minimizing pair has a positive-dimensional center. This proves the universal assertion in Proposition 5.1, stronger than existence of one point-centered minimizing pair.

§5.4; (5.17)–(5.21) · pp. 17–19

4. Proof of Lemma 6.2 — Making equality fix the entire profile

Proof target : Lemma 6.2 · pp. 21–22

Lemma 6.2 — The equality divisor and adjacent components

Suppose P is not generated at x , and fix t, m and a minimizing basis as in Proposition 5.1. Resolve $D_i = \text{div}(s_i)$ by $\pi : Y \rightarrow X$. Let E_j be the SNC support components, $a_j = A(E_j)$, and $z_j = (\text{ord}_{E_j} D_i / a_j)_i$. For a stratum Z whose image contains x , put $C_Z = \text{conv}\{z_j : j \in J(Z)\}$.

The vectors $q^0 \in \mathbf{Q}_{\geq 0}^{N_m}$ and $b^0 \in \mathbf{Q}_{> 0}^{N_m}$ supplied by Lemma 6.1 satisfy

$$b^0 \cdot q^0 = 1, \quad m \sum_i b_i^0 < n + 1, \quad b^0 \cdot z \leq 1 \quad (z \in C_Z, q^0 \in C_Z),$$

and q^0 belongs to no polytope indexed by a stratum with positive-dimensional image. Set $I = \{i : q_i^0 > 0\}$. Choose a positive integer M with all M/q_i^0 integral, and put $\mathfrak{c} = \sum_{i \in I} \mathcal{O}_X(-(M/q_i^0)D_i)$. Choose u with $\mathfrak{c}(uL)$ globally generated. On a principalization $\rho : T \rightarrow X$, retain a prime F_0 realizing q^0 and write $\mathfrak{c}\mathcal{O}_T = \mathcal{O}_T(-G)$. Make the pullbacks and exceptional divisors SNC, and for a sufficiently small rational $0 < \eta < 1$ define

$$B = (1 - \eta)\rho^* \sum_i b_i^0 D_i + \frac{\eta}{M} G.$$

For a prime F in the support, set $a_F = A(F)$ and $r_F = \sum_j \text{ord}_F(E_j) a_j \leq a_F$. Let S be the reduced sum of components satisfying $x \in \rho(F)$, $(\text{ord}_F(D_i) / a_F)_i = q^0$ and $r_F = a_F$. It contains F_0 , so $S \neq 0$.

Then every component of S maps to x and has coefficient a_F in B . Every other prime in the support meeting S has coefficient strictly smaller than its log discrepancy.

§6.1; Lemma 6.1; (6.1)–(6.5) · pp. 19–20 · §6.2; (6.6)–(6.9) · p. 21 · Lemma 6.2 · pp. 21–22

Proof route. A positive supporting functional alone does not fix the profile when equality holds. Perturb it using an ideal whose order is a minimum of coordinate ratios, and track the equality conditions in the resulting convex combination.

1 Rationalize positive supporting data

$$b^0 \cdot q^0 = 1, \quad b_i^0 > 0, \quad m \sum_i b_i^0 < n + 1$$

Preserve polytope membership and avoid positive-dimensional images

Retain the divisor F_0 of the rational ray and principalize the ideal.

[§6.1; Lemma 6.1; \(6.1\)–\(6.5\) · pp. 19–20](#)

[§6.2; \(6.6\)–\(6.9\) · p. 21](#)

2 Perturb the boundary by an ideal minimum

$$B = (1 - \eta)\rho^* \sum_i b_i^0 D_i + \frac{\eta}{M} G$$

$$\frac{\text{ord}_F G}{M} = \min_{i: q_i^0 > 0} \frac{\text{ord}_F D_i}{q_i^0}$$

Compare a positive weighted average with its minimum, including zero coordinates.

[\(6.10\)–\(6.12\) · p. 22](#)

3 Equality fixes the entire profile

$$\frac{\text{coeff}_F B}{a_F} = \frac{r_F}{a_F} ((1 - \eta)b^0 \cdot q' + \eta\mu) \leq 1$$

$$\text{Equality} \implies r_F = a_F, \quad q' = q^0$$

Equality forces the defining conditions of S ; other adjacent components are strict.

[Lemma 6.2 · pp. 21–22](#)

4 Obtain the configuration for local lifting

$$S \neq \emptyset, \quad \rho(S) = \{x\}$$

$$\text{coeff}_F B = a_F \quad (F \subset S), \quad \text{coeff}_F B < a_F \quad (F \not\subset S, F \cap S \neq \emptyset)$$

Combine rational support data with an ideal minimum to isolate the equality profile

[Open figure ↗](#) [TeX source ↓](#)

1. Rationalize support conditions and point-center constraints together

On each polytope containing the minimizing profile $q = (v(s_i)/A(v))_i$, the gradient gives $b \cdot z \leq b \cdot q = 1$. If q also belonged to a positive-dimensional-image polytope, it would realize a minimizing pair with a nonpoint center, contradicting Proposition 5.1. Lemma 6.1 chooses a rational point in the intersection of the smallest faces containing q , while avoiding the finitely many polytopes not containing it. After obtaining q^0 , it chooses $b^0 > 0$ in a rational feasible polyhedron while preserving $m \sum_i b_i^0 < n + 1$.

Toroidal extraction of the rational monomial ray gives F_0 . Its profile is q^0 , its retraction discrepancy inequality is an equality, and its center is x . This divisor is retained in subsequent resolutions.

[§6.1; Lemma 6.1; \(6.1\)–\(6.5\) · pp. 19–20](#) [· §2.1; Lemmas 2.1–2.2 · pp. 3–4](#)

2. Mixing in a minimum preserves the positivity needed for vanishing

The order of a sum of ideals is the minimum of their orders, hence

$$\frac{\text{ord}_F(G)}{M} = \min_{i \in I} \frac{\text{ord}_F(D_i)}{q_i^0}.$$

Meanwhile $u\rho^*L - G$ is globally generated. Since $D_i \sim mL$,

$$(n+1)\rho^*L - B \equiv \left(n+1 - (1-\eta)m \sum_i b_i^0 - \frac{\eta u}{M} \right) \rho^*L + \frac{\eta}{M} (u\rho^*L - G).$$

For small η the first coefficient is positive, the second summand is nef, and the total is nef and big. A component of S has retraction profile q^0 . Its retraction center has image containing x ; exclusion of positive-dimensional-image polytopes forces every such component to map to x . Both the minimum above and the pairing with b^0 equal a_F , giving $\text{coeff}_F B = a_F$.

§6.2; (6.6)–(6.9) · p. 21 · (6.10)–(6.12) · p. 22

3. Equality on an adjacent component forces membership in S

If F meets a component F_S of S , both retraction profiles lie in the same C_Z determined at an intersection point. For $r_F = 0$, all section orders and the order of G vanish, so $\text{coeff}_F B = 0 < a_F$. For $r_F > 0$, put $q' = (\text{ord}_F(D_i)/r_F)_i$ and $\mu = \min_{i \in I} q'_i/q_i^0$. Then

$$\mu \leq \sum_{i \in I} b_i^0 q'_i \leq b^0 \cdot q' \leq 1, \quad \frac{\text{coeff}_F B}{a_F} = \frac{r_F}{a_F} ((1-\eta)b^0 \cdot q' + \eta\mu) \leq 1.$$

Equality forces $r_F = a_F$ and $b^0 \cdot q' = \mu = 1$. The positively weighted average on I equals its minimum, so $q'_i = q_i^0$ there. Strict positivity of b_i^0 outside I also forces $q'_i = 0$ there. Thus $q' = q^0$. Since the intersection point lies over x , the prime F satisfies every defining condition for S . All adjacent components outside S therefore have strict inequality. This local comparison feeds into the main theorem's lifting step.

(6.10)–(6.12) · p. 22

5. Proof of Corollary 6.3 — Increasing the exponent by taking a product

Proof target : Corollary 6.3 · p. 23

Proof route. Since L need not have sections, global generation cannot simply be propagated by multiplying by another copy of L . Instead apply the all-dimensional Theorem 1.1 to a product with projective space.

1 Increase the product dimension as needed

$$m \geq n + 1, \quad r = m - n - 1, \quad Y = X \times \mathbf{P}^r$$

$$A = L \boxtimes \mathcal{O}_{\mathbf{P}^r}(1), \quad \dim Y = n + r$$

The all-dimensional theorem gives global generation of this adjoint.

[Theorem 1.1 · p. 1](#)

2 Apply the main theorem to the product

$$K_Y + (n + r + 1)A = (K_X + mL) \boxtimes \mathcal{O}_{\mathbf{P}^r}(n)$$

The smooth, connected, projective and ample hypotheses hold

Restriction preserves global generation; L itself need not be generated.

[Corollary 6.3 · p. 23](#)

3 Restrict to a fiber to obtain the required exponent

$$((K_X + mL) \boxtimes \mathcal{O}_{\mathbf{P}^r}(n))|_{X \times \{z\}} \simeq K_X + mL$$

Holds for every such integer m ; $r=0$ is the main theorem itself.

Increase the dimension, apply the main theorem, and restrict to one fiber

[Open figure ↗](#) [TeX source ↓](#)

1. Adjust the dimension and adjoint bundle together

Set $r = m - n - 1 \geq 0$, $Y = X \times \mathbf{P}^r$, and $A = L \boxtimes \mathcal{O}_{\mathbf{P}^r}(1)$. Then Y is smooth, connected and projective over $\mathbf{C}$, with $\dim Y = n + r$, and A is ample. Theorem 1.1 applied to (Y, A) makes

$$K_Y + (n + r + 1)A = (K_X + mL) \boxtimes \mathcal{O}_{\mathbf{P}^r}(n)$$

globally generated.

[Corollary 6.3 · p. 23](#)

2. Restrict to a fiber to recover every m

The restriction of a globally generated line bundle is globally generated. Restricting to $X \times \{z\}$ therefore gives global generation of $K_X + mL$. The case $r = 0$ is Theorem 1.1 itself. Since $m \geq n + 1$ was arbitrary, this proves the entire conclusion of Corollary 6.3.

[Corollary 6.3 · p. 23](#)

6. Which papers supply which steps

Source and result

Singularity exponents [DK] · Theorem 3.1; Lemma 3.2 · pp. 14–15

Used at

Lemma 2.3, §3, Lemma 4.3

Role and required hypotheses

Local semicontinuity; $e^\varphi = (\sum |g_i|^2)^{1/2}$ is locally Hölder. The manuscript adds constructibility through resolution and stratification

Checking scope

Compared the specified version and its use for closed threshold loci, degeneration and flags

Source and result

Kawamata–Viehweg vanishing [KV] · Theorem 0.1 · p. 1

Used at

Theorem 2.4, §3, §6.3

Role and required hypotheses

Smooth projective model, SNC fractional boundary and a nef and big remainder; H^1 vanishing gives surjective restriction

Checking scope

Compared the rounding formulation and the conditions at both lifting steps

Source and result

Brunn–Minkowski [BM] · Theorem 4.1 · print p. 362 / PDF p. 8

Used at

§5.3 (5.13)

Role and required hypotheses

Unit-cube thickenings of nonempty finite lattice sets; the sets and their Minkowski sum are bounded and measurable

Checking scope

Compared the inequality and its discrete application

Source and result

SNC resolution [RES] · Theorem 1.0.1 · print p. 781 / PDF p. 3; Definition 2.1.3 · print p. 783 / PDF p. 5; Theorem 2.4.1 · print p. 786 / PDF p. 8

Used at

§2.1, Proposition 4.4, §6.2

Role and required hypotheses

Characteristic zero, resolution and principalization preserving the SNC boundary; extracted primes retained by strict transform

Checking scope

Compared principalization and marked-ideal resolution statements with their use

Adjoint lifting [FL] is an explicit **methodological reference** in §6.3. Its threefold freeness theorem is not applied in higher dimension. The manuscript gives its own restriction argument on reduced $\mathcal{S}$ and descent with exceptional correction. For initial-exponent counting and the monomial threshold, the account follows the arguments provided in the manuscript; references to related theories are not counted as additional direct inputs.

Adjoint lifting [FL] · §1(1.6) · p. 3

7. Return to the sources

- Proposition 3.2 · pp. 6–9 — The equality case behind strictness and lifting from the point blowup.
- Lemmas 4.2–4.3; Proposition 4.4 · pp. 10–12 — Attainment with both bases and models varying.
- Lemma 5.2; (5.1)–(5.3) · pp. 13–14 — A restriction-image lower bound independent of the degree and singularities of the center.
- §5.4; (5.17)–(5.21) · pp. 17–19 — The fixed parameter, subsequence and order of limits.
- §6.1; Lemma 6.1; (6.1)–(6.5) · pp. 19–20 — Rationalization preserving support inequalities and the positivity budget.
- Lemma 6.2 · pp. 21–22 — The equality conditions forcing the same profile.
- §6.3; (6.13)–(6.15) · pp. 22–23 — Lifting using local effectivity and descent with exceptional positive correction.
- Corollary 6.3 · p. 23 — The product argument giving every higher exponent.

The proof text in §§2–6 was read. The statements and application sites of the four external inputs above were compared, as was the methodological connection to Fujita’s local lifting argument. The explanation tracks the nonzero section required by Theorem 1.1, the universal quantifier in Proposition 5.1, the three conclusions of Lemma 6.2 and the extension to every exponent in Corollary 6.3.

This is not an independent verification of the entire proof. The lattice approximation and choice of initial forms in §3, and all the asymptotic estimates in §5, were read as parts of the manuscript’s argument; a complete independent reconstruction or formal verification was not performed. Complete proofs of the external theorems and their further dependencies are outside the checking scope. All cross-catalogue dependencies also remain uninvestigated.

§§2–6 · pp. 3–23